# ScheduleSkies

# Software Requirements Specification

# Version 01.00

# Revision History

| Date | Version | Description | Author |
|---|---|---|---|
| 02/19//2026 | 01.00 | Initial Creation | Labiste, Medalla, Pescadero, Susan, Villamor |

# Table of Contents


1. Introduction 4
   - 1.1 Purpose 4
   - 1.2 Scope 4
   - 1.3 Definitions, Acronyms, and Abbreviations 5
   - 1.4 References 5
   - 1.5 Overview 5

2. Overall Description 6

3. Specific Requirements 6
   - 3.1 Functionality 6
     - 3.1.1 Smart Schedule & Itinerary Planner 6
     - 3.1.2 Weather-Aware Planning 7
     - 3.1.3 Traffic & Travel Prediction 7
     - 3.1.4 Real-Time Location Awareness 7
     - 3.1.5 Smart Place & Destination Discovery 7
     - 3.1.6 User Profiles & Personalization 8
     - 3.1.7 Smart Recommendations 8
     - 3.1.8 Notifications and Alert 8
     - 3.1.9 Collaboration and Sharing 8
     - 3.1.10 Map & Visualization Features 9
   - 3.2 Usability 9
     - 3.2.1 Graphical User Interface 9
     - 3.2.2 Accessibility 9
   - 3.3 Reliability & Availability 9
     - 3.3.1 Back-end Internal Computers 9
     - 3.3.2 Service Availability and Network 10
   - 3.4 Performance 10
     - 3.4.1 Response Time 10
     - 3.4.2 External Data Integration 10
     - 3.4.3 Resource Utilization 10
   - 3.5 Security 11
     - 3.5.1 Authentication and Authorization 11
     - 3.5.2 Data Protection 11
     - 3.5.3 Privacy and Data Access 12
     - 3.5.4 Web-Specific Security 12
   - 3.6 Design Constraints 12
     - 3.6.1 Standard Web Technologies 12
     - 3.6.2 Database and Cloud Constraints 12
     - 3.6.3 Operational Constraints 13
   - 3.7 Interfaces 13
     - 3.7.1 User Interfaces 13
     - 3.7.2 Hardware Interfaces 13
     - 3.7.3 Software Interfaces 14
     - 3.7.4 Communications Interfaces 14

# Software Requirements Specification

## 1. Introduction

The introduction of this Software Requirements Specification (SRS) provides a high-level view of the Schedule Skies platform. This document serves to gather and analyze the project's requirements to provide an in-depth insight into the system's capabilities. It defines the high-level features required to assist travelers, tourists, and event planners in choosing the best time and route through environmental analysis.

### 1.1 Purpose

The purpose of this document is to collect and analyze the assorted ideas required to define the Schedule Skies system and its requirements with respect to its consumers. This SRS predicts and outlines how the product will be used to gain a better understanding of the project, including concepts like real-time weather-aware planning and traffic prediction.

- This document provides a detailed overview of the software product, its parameters, and its goals.
- It describes the project's target audience, user interface, hardware, and software requirements.
- It defines how the client, development team, and audience perceive the product and its core functionalities.
- It serves as a guide for designers and developers throughout the Software Delivery Lifecycle.

### 1.2 Scope

The scope of this SRS pertains to the core features required to make the Schedule Skies project live, focusing on stakeholders and applications that allow for smart travel planning and event scheduling.

- The software focuses on analyzing environmental conditions, such as weather forecasts and traffic predictions, to assist users in choosing optimal times and routes.
- This SRS specifies the requirements for the software to be developed and can be used to

assist in the selection of internal or commercial software products.

- It serves as a model for defining organization-specific standards for travel-based web applications.

It focuses on the integration of location-based recommendations and collaboration tools for group planning.

### 1.3 Definitions, Acronyms, and Abbreviations

| Term | Definition |
| --- | --- |
| Itinerary | A daily or multi-day schedule or timeline for events and travel plans. |
| AI | Artificial Intelligence, utilized for suggested itinerary improvements and travel plan generation |
| GPS | Global Positioning System, used for real-time location tracking |
| Weather Suitability | A rating system for determining if outdoor activities are appropriate based on current forecasts. |

### 1.4 References

The requirements in this document are based on the following:

- Schedule Skies Project Overview.
- Schedule Skies Feature List.
- SRS Sample Template.

### 1.5 Overview

The following sections of this document provide a comprehensive description of the system.

- **Section 2** discusses the overall description of the project, user characteristics, and constraints.
- **Section 3** provides the specific requirements, including functional needs like weather-aware planning and performance standards.
- **Section 3.7** details the external interface requirements, such as interactions with map and booking platforms.

## 2. Overall Description

This document describes the proposed system, **Schedule Skies**, a smart web-based event and travel planning application. The current challenge in travel and event planning is the lack of integration between scheduling, weather forecasting, traffic prediction, and location-based recommendations. Travelers and event planners often rely on multiple separate applications, resulting in inefficient planning, delays, safety risks, and scheduling conflicts.

Schedule Skies aims to solve this problem by providing a unified platform that dynamically optimizes itineraries based on real-time environmental data, traffic data, user preferences, and location awareness. The system ensures smoother, safer, and more efficient travel and event experiences.

## 3. Specific Requirements

This section details the functional, design, usability, reliability, performance, security, and supportability requirements for the ScheduleSkies application.

### 3.1 Functionality

This subsection contains the functional requirements of Schedule Skies. These requirements are organized based on the system's core features. Each feature will be refined into use cases, diagrams, and traceability matrices to ensure completeness and clarity.

#### *3.1.1 Smart Schedule & Itinerary Planner*

3.1.1.1 The system shall allow users to create daily schedules.

3.1.1.2 The system shall allow users to create multi-day itineraries.

3.1.1.3 The system shall recommend adjustments to schedule timing based on weather conditions.

3.1.1.4 The system shall detect conflicts such as overlapping events.

3.1.1.5 The system shall notify users of unrealistic travel times.

3.1.1.6 The system shall allow users to resolve scheduling conflicts.

#### *3.1.2 Weather-Aware Planning*

3.1.2.1 The system shall provide weather forecasts per location.

3.1.2.2 The system shall display weather-based alerts (rain, storms, heat, floods).

3.1.2.3 The system shall provide weather suitability ratings for outdoor activities.

3.1.2.4 The system shall suggest alternative dates or times if weather conditions are unfavorable.

#### *3.1.3 Traffic & Travel Prediction*

3.1.3.1 The system shall estimate travel time between selected destinations.

3.1.3.2 The system shall suggest traffic-aware routes.

3.1.3.3 The system shall provide travel delay warnings.

#### *3.1.4 Real-Time Location Awareness*

3.1.4.1 The system shall track the user's current location using GPS.

3.1.4.2 The system shall suggest nearby places based on the user's current location.

#### *3.1.5 Smart Place & Destination Discovery*

3.1.5.1 The system shall allow users to search for restaurants, hotels, attractions, and events.

3.1.5.2 The system shall allow filtering by budget.

3.1.5.3 The system shall allow filtering by distance.

3.1.5.4 The system shall allow filtering by ratings.

3.1.5.5 The system shall allow filtering by weather suitability.

3.1.5.6 The system shall display availability of information, including open hours and peak times.

3.1.5.7 The system shall notify users if a selected place is closed or unavailable.

#### *3.1.6 User Profiles & Personalization*

3.1.6.1 The system shall allow users to create a profile.

3.1.6.2 The system shall store saved itineraries and favorite locations.

3.1.6.3 The system shall track travel history and analytics.

3.1.6.4 The system shall authenticate user credentials.

3.1.6.5 The system shall allow users to update their profile information.

#### *3.1.7 Smart Recommendations*

3.1.7.1 The system shall provide AI-suggested itinerary improvements based on choices.

3.1.7.2 The system shall suggest alternative locations if weather or traffic conditions become unfavorable.

#### *3.1.8 Notifications and Alert*

3.1.8.1 The system shall send event reminders.

3.1.8.2 The system shall notify users of sudden weather changes affecting their itinerary upon logging in.

#### *3.1.9 Collaboration and Sharing*

3.1.9.1 The system shall allow users to share itineraries with others.

3.1.9.2 The system shall support group planning features.

3.1.9.3 The system shall allow users to vote on places and schedules.

#### *3.1.10 Map & Visualization Features*

3.1.10.1 The system shall provide interactive maps.

3.1.10.2 The system shall display heatmaps for traffic conditions.

3.1.10.3 The system shall provide a timeline and map hybrid view.

### 3.2 Usability

#### *3.2.1 Graphical User Interface*

3.2.1.1 The system shall provide a consistent look and feel across all web pages.

3.2.1.2 The system shall provide an interactive and user-friendly interface.

3.2.1.3 The system shall use icons, toolbars, and visual indicators for navigation.

3.2.1.4 The system shall provide map-based and timeline-based visual representations.

#### *3.2.2 Accessibility*

3.2.2.1 The system shall be responsive across desktop and mobile devices.

### 3.3 Reliability & Availability

#### *3.3.1 Back-end Internal Computers*

The system shall ensure data integrity through the use of Supabase's built-in database constraints and row-level security.

The system shall rely on Supabase’s automated backup and point-in-time recovery (PITR) to ensure data is replicated and restorable in case of failure

The system shall store all user itineraries and profile data in a Supabase database.

#### 3.3.2 Service Availability and Network

The system shall provide a contractual agreement with the cloud provider (Supabase) to ensure 99.9% or higher service availability.

The system shall ensure that the database remains reachable via secure API calls even during periods of high user density on the web app.

### 3.4 Performance

#### 3.4.1 Response Time

The system shall aim for a Largest Contentful Paint (LCP) of less than 2.5 seconds to ensure the user feels the app is immediately available.

Utilizing Supabase's PostgREST and indexing, simple read operations (e.g., viewing an itinerary) shall return results in under 500ms under normal network conditions.

The system shall acknowledge user interactions (button clicks, navigation) within 200ms to maintain a sense of instantaneous response.

#### 3.4.2 External Data Integration

The system shall fetch real-time weather and traffic data asynchronously so that the main UI remains interactive while data is being retrieved.

Interactive map elements shall be optimized for smooth panning and zooming at 60 frames per second (16.7ms per frame).

#### 3.4.3 Resource Utilization

The web app shall be optimized for low data consumption, ensuring that travelers with limited or

roaming data plans can still access their schedules effectively.

Background processes (like real-time notification polling) shall be optimized to minimize impact on mobile device battery life.

## 3.5 Security

### 3.5.1 Authentication and Authorization

The web app shall use Supabase Auth for secure user registration and login, supporting email/password and potentially OAuth providers (e.g., Google) for convenience.

The system shall implement PostgreSQL RLS policies to ensure that a user can only View, Edit, or Remove itineraries that belong to their specific user_id. No user shall be able to access another user's private travel plans.

The web app shall use secure, HTTP-only cookies or protected local storage tokens to manage user sessions, with automatic expiration to prevent unauthorized access on shared computers.

### 3.5.2 Data Protection

The web app shall enforce HTTPS/TLS encryption for all communications between the user's browser and the Supabase servers to prevent "man-in-the-middle" attacks.

All user data, including account details and itinerary notes, shall be encrypted at rest within the Supabase managed database.

The system shall never store plain-text passwords; all passwords shall be hashed and salted using industry-standard algorithms (handled by Supabase Auth).

#### 3.5.3 Privacy and Data Access

The system shall only request the user's geolocation for map and traffic functions with explicit browser-level permissions.

The system shall provide an "Account Deletion" feature that allows users to permanently remove all their data from the Supabase database in compliance with data privacy regulations.

#### 3.5.4 Web-Specific Security

The web app shall sanitize all users in the "Add Event" forms to prevent malicious scripts from being executed.

The backend (Supabase) shall be configured to only allow requests from the official Schedule Skies web domain to prevent unauthorized API access from other websites.

### 3.6 Design Constraints

#### *3.6.1 Standard Web Technologies*

- The system shall be built as a travel-based web application.
- The system focuses on stakeholders and applications that allow for smart travel planning and event scheduling

#### *3.6.2 Database and Cloud Constraints*

- The system shall store all user itineraries and profile data in a Supabase database.
- The system shall ensure data integrity through the use of Supabase's built-in database constraints and row-level security.
- The system shall rely on Supabase's automated backup and point-in-time recovery

(PITR) to ensure data is replicated and restorable in case of failure.

- The system shall provide a contractual agreement with the cloud provider (Supabase) to ensure 99.9% or higher service availability.

#### *3.6.3 Operational Constraints*

- The system focuses on analyzing environmental conditions, such as weather forecasts and traffic predictions, to assist users in choosing optimal times and routes.
- The system shall ensure that the database remains reachable via secure API calls even during periods of high user density on the web app

### 3.7 Interfaces

There are multiple interfaces supported by the Schedule Skies software system to ensure seamless environmental analysis and scheduling.

#### *3.7.1 User Interfaces*

- The user interface must be accessible via standard web browsers to assist travelers, tourists, and event planners in choosing the best time and route.
- The user interface shall support the integration of location-based recommendations and collaboration tools for group planning.

#### *3.7.2 Hardware Interfaces*

- The system shall interface with a Global Positioning System (GPS) for real-time location tracking.

- Since the software relies on real-time weather-aware planning and traffic prediction, hardware must have an active internet connection.

#### *3.7.3 Software Interfaces*

- The Schedule Skies system shall include interactions with map and booking platforms.
- The system shall interface with external weather data providers to determine weather suitability and generate a rating system for outdoor activities.
- The system shall communicate with Artificial Intelligence (AI) algorithms for suggested itinerary improvements and travel plan generation.

#### *3.7.4 Communications Interfaces*

- The system shall utilize secure API calls to communicate between the web application and the Supabase database.